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Agricultural water management practices to improve the climate resilience of irrigated agriculture in India

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Abstract

The projected implications of climate change for water and agriculture to meet diverse and competitive water demands requires smart water management solutions. Science- and evidence-based, agricultural water management (AWM) can significantly contribute to reduce unsustainable water use and help enhance water resilience and adaptation to climate change. This paper presents a brief overview of potential AWM practices focusing on enhancing water resilience, increasing yields, and wherever possible, reducing emissions. This is achieved via increased land and water use efficiency, water and energy savings, and improved water productivity with considerable scope to improve agricultural resilience. In this context, the prioritization of a location-specific portfolio of smart AWM practices to make the right investment decisions is very important. We present two distinct and complementary approaches to prioritize AWM practices in this paper: one follows stakeholder analysis to build a prioritized portfolio of climate-smart AWM practices and the other employs a simple water balance-based approach to prioritize interventions. The way forward in mainstreaming and scaling out context-specific climate-smart AWM interventions is also discussed with a focus on capacity building, water management extension services, and the mobilization of resources through the convergence of institutions and co-financing from relevant development schemes.

KEYWORDS

agricultural water management, climate resilience, climate-smart agricultural water management practices, convergence, irrigation

Résumé

Les implications prévues du changement climatique pour l'eau et l'agriculture pour répondre à des demandes en eau diverses et compétitives nécessitent des solutions intelligentes de gestion de l'eau. La gestion de l'eau agricole, fondée sur des données scientifiques et factuelles, peut contribuer de manière significative à réduire l'utilisation non durable de l'eau et aider à renforcer la résilience de l'eau et l'adaptation au changement climatique. Ce document présente un bref aperçu des pratiques potentielles de gestion de l'eau agricole

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(AWM) axées sur l'amélioration de la résilience de l'eau, l'augmentation des rendements et, dans la mesure du possible, la réduction des émissions. Ceci est réalisé grâce à une efficacité accrue de l'utilisation des terres et de l'eau, des économies d'eau et d'énergie et une amélioration de la productivité de l'eau avec une portée considérable pour améliorer la résilience agricole. Dans ce contexte, la priorisation d'un portefeuille de pratiques GEA intelligentes spécifiques à un lieu pour prendre les bonnes décisions d'investissement est très importante. Nous présentons deux approches distinctes et complémentaires pour hiérarchiser les pratiques de GEA dans cet article: l'une suit l'analyse des parties prenantes pour construire un portefeuille hiérarchisé de pratiques de GEA intelligentes face au climat, et l'autre utilise une approche simple basée sur le bilan hydrique pour hiérarchiser les interventions. La voie à suivre pour intégrer et étendre les interventions de GEA intelligentes face au climat et spécifiques au contexte est également discutée en mettant l'accent sur le renforcement des capacités, les services de vulgarisation de la gestion de l'eau et la mobilisation des ressources grâce à la convergence des institutions et au cofinancement des programmes de développement pertinents.

MOTS CLÉS

gestion de l'eau agricole, pratiques de gestion de l'eau agricole intelligentes face au climat, résilience climatique, convergence, irrigation

1 | BACKGROUND

With agriculture currently using 70% of freshwater resources globally and growing water demands from competing sectors, the food production system is increasingly exposed to the risks associated with water availability, quantity, and quality to support a growing population by 2050. Climate change is expected to intensify the existing risks, particularly in countries where water scarcity is already a concern (Iglesias & Garrote, 2015). The irrigated areas will increase in coming years, at the same time that more freshwater supplies will be diverted from agriculture to meet the increasing demands for water from the domestic, industrial, urban, and environmental sectors. In developing countries, where water extraction is often unregulated, unpriced, and subsidized, irrigated agriculture is placing increasing pressure on available freshwater resources (Fishman et al., 2015). This is particularly being felt in India, with a predominantly agrarian economy supporting about 18% of the global population with just 4.2% of the world's freshwater resources. Agriculture remains the predominant means of livelihood, employing over 50% of the Indian workforce and contributing about 17% to national gross domestic product (Department of Economic Affairs, 2019).

Out of the 140 Mha (million hectares) net cultivated area in India, nearly 48.8%, or 68.4 million Mha, is irrigated (Ministry of Agriculture and Farmers Welfare, 2017). The remaining 51.2% is rainfed agriculture. In India, agriculture consumes about 82% of total freshwater withdrawals, and increasing the area being irrigated puts pressure on both surface and groundwater resources. Because of competing water demands in the future, there is likely to be a reduction in the diversion of freshwater to agriculture, with some estimates predicting it at 72% by 2025 and 65%–68% in 2050 (Asian Development Bank [ADB], 2013). Climate change is now a reality, and its negative impacts may put regional water balance and current water management at significant risk (Intergovernmental Panel on Climate Change [IPCC], 2014). Growing water use as a result of an increasing population, economic development, changing lifestyles, and shifting consumption patterns, combined with a more erratic and uncertain water supply, will aggravate the situation of currently water-stressed regions and generate water stress in regions where water resources are still abundant today (UNESCO, UN-Water, 2020). As a result, the sustainable management of already stressed water resources and agriculture is a challenge. Agricultural water management (AWM) to enhance water resilience and adaptation to climate

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change is fundamental in order to sustain Indian agriculture.

2 | INDIA'S IRRIGATION GROWTH TRAJECTORY

Irrigation development in India has been a high priority since independence throughout the government's successive 5-year plans. Huge investments were made in the development of canal irrigation, but this investment decelerated around the 1970s, and neglect of canal irrigation gave way to groundwater irrigation. Figure 1 shows the growth of irrigation by source in India. The net irrigated area increased from nearly 18% in the early 1960s to about 48% in recent times (Jain et al., 2019). As evident from Figure 1, surface water from canals and tanks covered most irrigated lands till about the late 1960s and early 1970s. With the advent of modern drilling and pumping technologies, government policies through subsidies, and rural energy supplies, groundwater development received a big boost. The number of groundwater irrigation structures went up from 6.2 million in 1986–1987 to 20.5 million in 2013–2014 (Mukherji, 2020). Moreover, the area irrigated by groundwater has increased greatly: While it comprised only 29% of the total irrigated area in 1950–1951, today it accounts for 63% (Jain et al., 2019).

Groundwater-led irrigation was instrumental in the success of the Green Revolution in India, mostly because of its accessibility and reliability as compared to canal irrigation (Mukherji, 2020). Groundwater abstraction is also not uncommon in canal command areas through wells and tube wells and a case of conjunctive use, perhaps more by default than by design. Massive water harvesting and integrated watershed management programmes are also helping to provide limited irrigation to rainfed and dryland agricultural areas, at least for supplemental irrigation, either through micro-level surface water bodies/ponds and/or enhanced groundwater recharge to wells. Irrigation has been pivotal in India's agricultural growth and path to food security. Zaveri and Lobell (2019) showed that national wheat yields in the 2000s are 13% higher than they would have been without the irrigation trends that have arisen since 1970. Irrigation has also mitigated the impact of weather extremes to an extent. However, currently growth in irrigation has stagnated while also being plagued by number of challenges.

2.1 | Challenges in irrigated agriculture

2.1.1 | Gap between irrigation potential created and irrigation potential utilized

An important challenge facing the irrigation sector in India has been the growing gap between irrigation potential created (IPC) and irrigation potential utilized (IPU) (Figure 2a), mostly in surface water-irrigated systems. The total IPU at the end of the Eleventh Five Year Plan was 87.86 Mha against a total of 113.53 Mha IPC, a gap

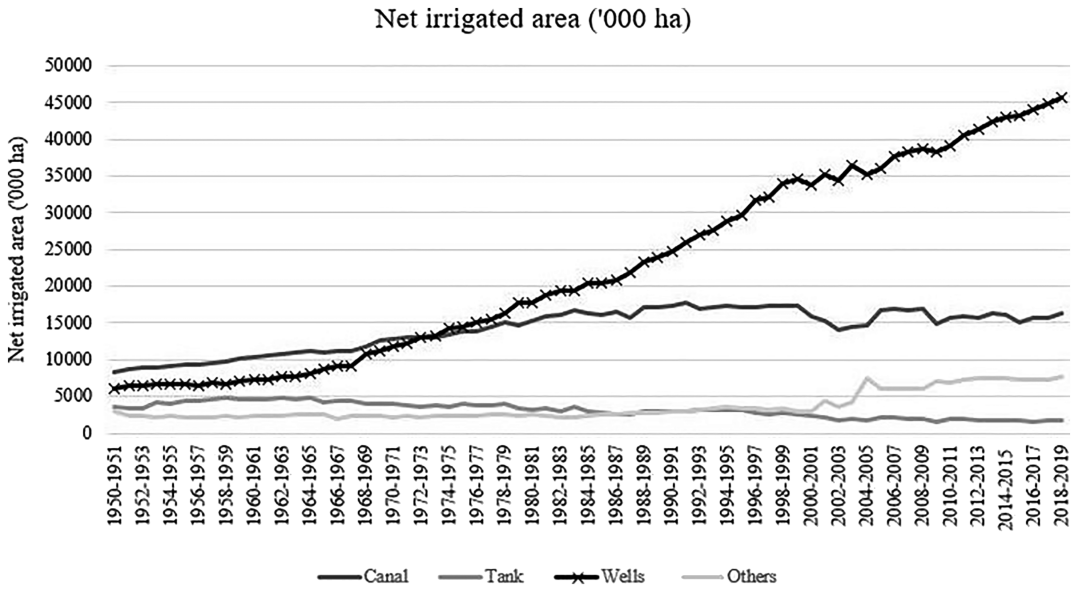


FIGURE 1 Net area irrigated ('000 ha) by irrigation source in India (1951–2015). *Source:* Ministry of Agriculture and Farmer Welfare, Govt. of India

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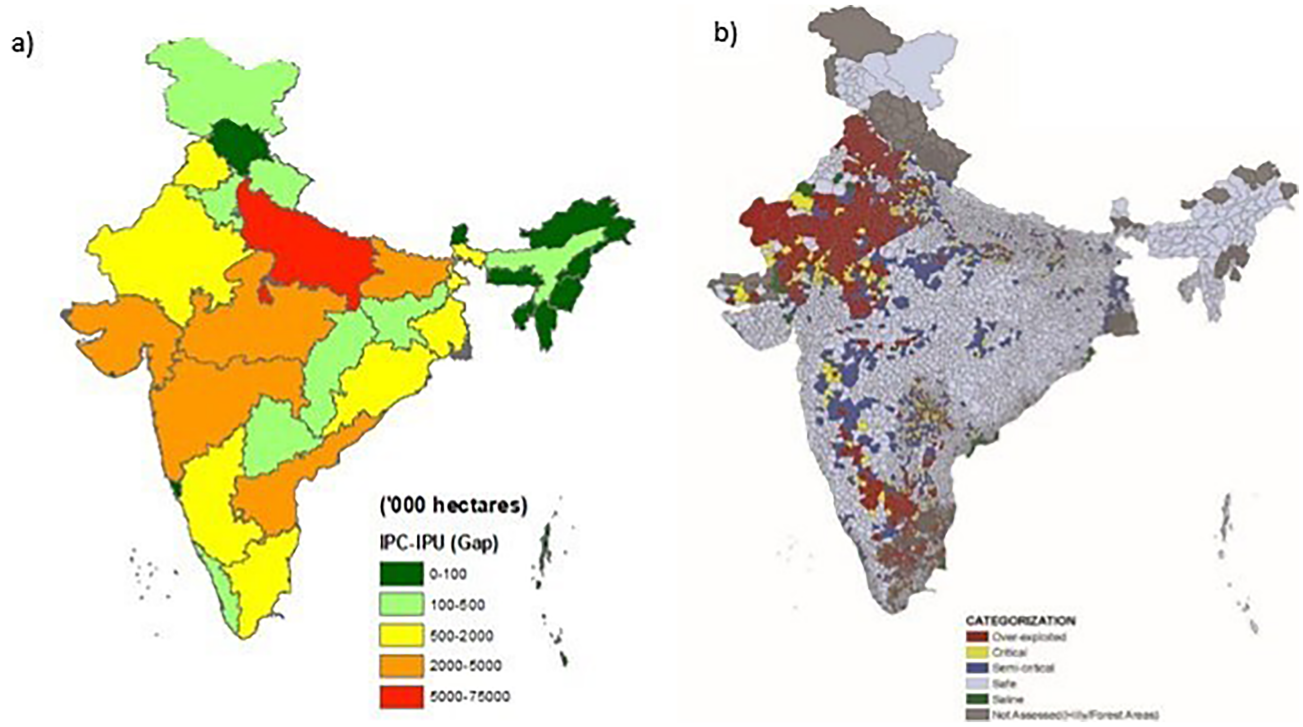


FIGURE 2 (a) Gap between irrigation potential created (IPC) and utilized (IPU) for Indian states at the end of 2013–2014 ('000 ha). *Source:* Central Water Commission (CWC), 2015. (b) Categorization of groundwater assessment units. *Source:* Central Ground Water Board (CGWB), 2019

of 25.67 Mha (Central Water Commission [CWC], 2015). Delays in developing command areas and on-farm water development (field channels, land levelling/shaping) were the key reasons for this non-utilization.

2.1.2 | Groundwater depletion

Groundwater has become a major source of irrigation, with 90% of withdrawal being used for irrigation and 63% of irrigated area using groundwater, leading to a serious risk of aquifers being overexploited. In India, out of the total 6,881 assessment units (blocks/mandals/talukas/*firkas*), 1,186 units in various states (17%) have been categorized as "overexploited," indicating that groundwater extraction exceeds the annually replenishable groundwater recharge. In addition, 313 units (5%) are "critical," with groundwater abstraction ranging between 90% and 100% of the recharge (Central Ground Water Board [CGWB], 2019) (Figure 2b). In India, unsustainable groundwater (UGW) abstraction produces enough food to feed 173 million people, so losing access to UGW will aggravate food security concerns (Zaveri et al., 2016). A recent study shows that without any adaptations, based on the current trajectory of groundwater depletion, cropping intensity may decrease by 20% in India and by 68% in groundwater-depleted regions (Jain et al., 2021). The critical role of groundwater in building resilience in South Asian agriculture, with particular emphasis on India, has been discussed in Sikka et al. (2020).

2.1.3 | Low irrigation efficiency and water productivity

Agriculture already accounts for more than 70% of freshwater withdrawals globally and up to 95% in developing countries. While more water will be required to meet future food demand, its availability is constrained by frequent water shortages, resource degradation, the variability of climate change, increasing population growth, and increasing competition for water from other sectors (Amarasinghe et al., 2010, 2021). This will require increasing water productivity, meaning "more crop or value per drop of water" for which there is large potential globally and in India (Alam et al., 2021; Amarasinghe et al., 2010; Sharma et al., 2018). This is critical in India, which is marred by low irrigation efficiencies and low water productivity (Sharma et al., 2018). According to a study conducted by the Central Water Commission (CWC), the average water use efficiency across 30 completed major and medium

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irrigation projects is only around 38%. An assessment of water productivity for 10 main crops at a national scale showed a large variation in water productivity, reflecting both the need and the potential to increase water productivity (Sharma et al., 2018). Their study also reported that the major water consuming crops, namely rice, wheat, and sugarcane, consume almost 80% of irrigation water. Paddy and sugarcane together occupy just one-fourth of the gross cropped area, while consuming over 60% of the total water used in irrigation. Thus, any water management strategy to enhance resilience and adaptation to climate change should have water productivity as its cornerstone.

2.1.4 | Vulnerability to climate change

Agriculture is highly vulnerable to weather extremes and climate change. This is exemplified by analysing agricultural production losses from natural disasters where more than 80% of the loss was due to extreme weather events such as floods and droughts (Food and Agriculture Organization of the United Nations [FAO], 2015). With extreme weather expected to increase in the future because of climate change, the vulnerability of agriculture is also likely to increase. There is already evidence that climate change is impacting agricultural growth. An analysis by Ortiz-Bobea et al. (2021) shows that since 1961, climate change has reduced global agricultural total factor productivity by 21%, which is equivalent to the last 7 years of productivity growth. The adverse impact of climate change on Indian agriculture is expected to reduce annual agricultural incomes on average by 15%–18% (Department of Economic Affairs, 2018). Though irrigating land will help mitigate some of the impact, against the backdrop of more frequent weather extremes, water scarcity, groundwater depletion, and competition from other sectors, sustaining and growing irrigation will be a challenge. This will require making irrigated agriculture more resilient.

With increasing risks from climate change, reduced diversion of freshwater to agriculture because of competing demands from other sectors, depleting groundwater resources in areas that are already water stressed, low water use efficiency and water productivity and stagnating crop yields, improving water resilience becomes high priority for sustainable and resilient agriculture. With water at the heart of adaptation to climate change, AWM is critical to building water resilience and adapting to climate change. The Global Commission on Adaptation (GCA) has also emphasized the importance of adaptation at the local level, with locally led planning and action, as more relevant in managing irrigation and AWM (Global Centre on Adaptation (GCA) and World Resources Institute (WRI), 2019). This paper presents a portfolio of evidence-based potential AWM interventions with a focus on enhancing water resilience, increasing yields and, wherever possible, reducing emissions. This can be achieved via increased land and water use efficiency, water and energy saving, and improving water productivity. Two distinct and complementary approaches to prioritize AWM practices are also presented: one based on stakeholder analysis and the other one employing a simple water balance-based approach. In order to scale out climate-smart AWM interventions, the paper suggests using convergence and co-financing to build capacity, provide water management extension services, and mobilize resources.

3 | AWM: A KEY TO CLIMATE-RESILIENT AGRICULTURE

As water is the medium of expression of the impacts of climate change, water management is at the centre of climate change adaptation (Sikka et al., 2018). Water is to adaptation what energy is to mitigation (United Nations, 2019). Climate-resilient agriculture refers to agricultural practices that enhance the capacity of the agroecosystem to relieve, resist, respond, and recover from climate-related disturbances. Conceptually, AWM is the management of water used in agriculture to enhance agricultural production with optimum water use. AWM has been defined in a variety of ways, meaning different things to different users. In a broader sense, AWM includes the management of water used in crop production (both rainfed and irrigated), livestock production, and inland fisheries (International Fund for Agricultural Development [IFAD], 2015). Water and land management is one of the major pathways to enhancing resilience and reducing greenhouse gas (GHG) emissions in agriculture as well. Smart AWM practices, when properly designed and implemented in the field, help increase surface, groundwater, and soil moisture availability, improve soil water balance, and reduce evaporation/seepage and non-beneficial losses to enhance crop yields/production, water use efficiency, and water productivity. In most cases, they also help reduce GHG emissions because of improved water, energy, and nutrient management. So, AWM interventions contribute to three key elements of climate-smart agriculture (CSA) by increasing yields/production, adapting and building resilience to climate change, and possibly reducing GHG emission (Aggarwal et al., 2018; Food and Agriculture Organization of the United Nations (FAO), 2010).

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3.1 | Potential AWM practices for building resilience

India has a long history of agricultural irrigation and water management. Many researchers in India have demonstrated how the promotion of innovative AWM interventions can build resilience to climate change (Anantha et al., 2020; Jat et al., 2020; Patle et al., 2019; Sikka et al., 2016, 2018). The Indian Council of Agricultural Research (ICAR), through its various institutions, and the All India Coordinated Research Project on Irrigation Water Management (AICRP-IWM), state agricultural universities (SAU), water and land management institutes (WALMI) in the states, the International Water Management Institute (IWMI), and other CGIAR (formerly the Consultative Group for International Agricultural Research) centres have contributed to bringing out area- and crop-specific improved AWM practices to increase water use efficiencies, water productivity, and to build resilience. The National Initiative on Climate-Resilient Agriculture (NICRA; renamed National Innovations in Climate-Resilient Agriculture in 2014) of the ICAR generated field evidence of enhanced resilience under its Technology Demonstration Component (TDC) through area-specific smart practices in farmers' participatory mode with a major focus on natural resources management in 121 climatically vulnerable districts across the country (Prasad et al., 2014, 2015).

AWM practices can be broadly categorized as water demand management and supply augmentation. Demand management involves reducing irrigation water demand (and potentially reducing non-beneficial evaporation and non-reusable return flows) through the implementation of interventions focusing on increasing irrigation efficiency (e.g., micro-irrigation, smart irrigation scheduling, supplemental irrigation, deficit irrigation), laser-guided land levelling (LLL), or reducing actual consumptive water demand by changing cropping pattern and production systems, introducing short duration varieties of seeds and changing sowing dates, manipulating seedling age, planting direct-seeded (DSR) or wet-seeded rice, integrating agriculture and aquaculture, increasing soil fertility, adding organic matter, tillage and soil mulching, conservation agriculture (CA), etc. Supply augmentation AWM interventions increase surface, subsurface, and groundwater storage either by using in situ soil and water conservation practices (e.g., mulching, field bunding, increasing bund height around rice fields) or ex situ storage interventions (e.g., rainwater harvesting, aquifer recharging) (Barron et al., 2009; Sikka et al., 2016, 2018).

A range of AWM interventions have been put into practice in Indian agriculture, though at varying levels and scales, and there is a need to scale these out to the development level in order to have transformational and visible impacts in building resilience to climate change. However, both in India and elsewhere, demand-side management measures have received far less attention compared to supply-side augmentation. In the following section, based on field evidence, we provide a brief account of some key AWM practices which are being promoted and/or have the potential to build resilience.

3.1.1 | Demand management

Demand management practices, by improving irrigation efficiency, not only save water but also reduce pumping hours, saving energy (electricity or diesel fuel) and thus leading to savings in associated costs, reduced subsidy burdens, and reduced carbon footprints and GHG emissions. Some of the key demand management interventions which are practiced at scale or have the potential to build resilience are presented here.

Increasing water application/use efficiency

The optimum utilization of available water resources and higher water use efficiency are critical in adapting to climate change. India's National Water Mission (one of the eight national missions under the National Action Plan for Climate Change) has the target of increasing water use efficiency by 20%. In the context of climate change, increasing water use efficiency is not only important for saving water but also for saving energy and reducing GHG emissions. Water demand management interventions such as LLL and pressurized irrigation systems (drip and sprinklers) are being promoted by the federal and state governments through various subsidized programmes to build resilience and increase water productivity.

Land levelling is helpful to maintain the uniform slope of the land surface, to conserve water during scanty rains, and to facilitate uniform water application in the field and root zones. It increases the application and distribution efficiencies of irrigation, which ultimately lead to higher crop yields and water savings and enhances water productivity. Surface irrigation on uneven and sloping farmlands is one of the major sources of applied irrigation water loss, resulting in low water application efficiency. While conventional and traditional land levelling has been commonly used, LLL is now being recommended and scaled up. LLL can level up to ±1 cm as compared to ± 5 cm in traditional land levelling. Field-based evidence has shown that laser-levelled land can save 20%–30% of applied irrigation water, thus increasing water application efficiency by up to 50% while at the

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same time improving crop yields by 10%–15% and reducing energy use by 15% (Aryal et al., 2015; Lohan et al., 2014). Laser land levellers were introduced in rice–wheat irrigated systems in north-western India in the early 2000s and quickly being adopted in both the north-west and other areas. Most farmers hire LLL services from private service providers or custom hiring centres.

Micro-irrigation is an effective water-, energy-, and nutrient-smart climate-resilient technology to save water and energy, increasing crop yields and reducing GHG emissions compared with flood/surface irrigation methods. Drip irrigation improves irrigation efficiency by providing water directly to crop roots and reducing water losses from surface evaporation, runoff, and deep percolation. Micro-irrigation systems have been receiving considerable attention from various government programmes in India, including Pradhan Mantri Krishi Sinchayee Yojana (PMKSY), for more crop per drop. While flood irrigation can achieve water use efficiency up to 65%, the use of sprinkler and drip irrigation could help achieve 85% and 90% water use efficiency, respectively (Howell, 2003). Field studies of drip irrigation systems have demonstrated water saving and enhanced water use efficiency by 35%–40% over conventional systems. Studies have also shown that micro-irrigation systems bring substantial benefits in terms of water saving, productivity gains, labour and energy savings, and reduced GHG emissions (Table 1) (Sikka et al., 2016). Even in a scenario where crop water requirements are likely to increase with increasing temperatures, reducing application losses and enhancing water use efficiency and crop water productivity will help to mitigate water deficits. Among different irrigation methods (surface, rain gun, sprinkler, and drip), the subsurface drip irrigation (SDI) system is the most efficient and has the lowest CO_2 emissions, but its adoption is still in the early stages in India. Early pilot programmes have shown the potential of SDI as a climate-smart intervention through irrigation water savings of 48%–53% in rice and 42%–53% in wheat compared to flood irrigation while maintaining the same yields (Sidhu et al., 2019). This is emerging as a promising technology for climate-resilient agriculture, especially in water-deficient and groundwater-irrigated areas.

Underground pipeline (UGPL)/piped water distribution systems are another potential water-smart intervention that has been receiving attention. UGPL efficiently conveys tube well or canal water using irrigation pipes and is useful in undulating or poorly levelled sandy soils where water losses are high and the elevated land cannot be irrigated with open channels. UGPL minimizes water losses during conveyance from the water source to farmers' fields.

Direct-seeded rice

Rice is one of the most important cereal food crops of India and is grown under diverse soil and climatic conditions. The area growing rice was 43.9 million ha during 2014–2015, which translates to a little more than one-fifth of India's total cropped area. Rice is highly water intensive, and its cultivation in the semi-arid north-western states and in other groundwater-irrigated areas has led to large-scale groundwater depletion. Reducing demand through better AWM practices is thus critical to make food systems resilient and sustainable. Puddled transplanted rice and continuous flooding are the traditional practices of these states. DSR with no or reduced tillage is a feasible alternative to conventional puddled transplanted rice with the potential to save water, energy, and labour and reduce GHG emissions (Pathak et al., 2011). DSR saves water by avoiding puddling, transplanting, and standing water. DSR can save up to 12%–33% of the irrigation water used for raising nursery and puddling (Kumar & Ladha, 2011). DSR maintains similar yield levels with less irrigation water and lower global warming potential (GWP). DSR technology was recommended for the north-western states of Punjab and Haryana, the rice bowl of India and the centre of groundwater depletion, during the 2010s, and consequently the area using DSR has increased steadily. However, its large-scale adoption has been constrained by concerns of weed infestation, though scientists have recommended herbicides to tackle this issue. It is worth noting that during the COVID-19 pandemic, large numbers of farmers in Punjab, for instance, switched to DSR during kharif 2020 because of labour scarcity as a result of migration. Thus, the area growing DSR increased to an all-time high, 34% higher than the total area of the last decade (Chaba, 2020). There is field-based evidence to support DSR's potential not only as an important climate-resilient intervention (Table 1) but also to help farmers grow rice during labour shortages as seen in Punjab.

Alternate wetting and drying method of irrigation

Another recommended irrigation alternative for irrigated rice systems is alternate wetting and drying (AWD). Under AWD, rice fields are subjected to alternate cycles of saturated and unsaturated conditions where irrigation is interrupted and water is allowed to subside until the ponded water disappears and the soil reaches a certain moisture level (Carrijo et al., 2017). AWD can reduce irrigation water inputs by 23%–43% and can reduce the GWP by 45%–90% as compared to continuously flooded rice (Adhya et al., 2014). Pathak (2015) reported a reduction in GWP by 25%–30% using intermittent flooding in rice over continuous flooding.

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TABLE 1 Savings in consumptive water use, irrigation water, and energy; improvement in crop yield and other co-benefits for improved water management practices for the rice–wheat food system in Punjab and Haryana

Water management practice/ technology/policy	Reference	Consumptive water use saving (ET)	Irrigation water saving (L)	Energy savings (%)	Yield improvement (%)	Environmental and other co-benefits
1. Direct-seeded rice	Kakraliya et al. (2018) Sidhu et al. (2019)	10%–15% over transplanted rice	30% over transplanted rice	~ 15	+ 10	Helps in managing stubbles
2. Laser-guided land levelling	Lohan et al. (2014) Aryal et al. (2015)	2%[a]	20%–30%	10–15	10–15	Smaller C footprint
3. Installation of UGPL systems	DSWC, Punjab	–	30% compared to open channel	40	~ 10–20	Benefits to tail end command, labour savings
4. Conservation agriculture	Kakraliya et al. (2018)	–	14%–30%	25–30	3–16	Increased net income, increased fertilizer efficiency
5. AWD method of irrigation	Adhya et al. (2014) Pathak (2015) Carrijo et al. (2017) Upadhyaya (2018)	< 5%	23%–43%	~ 5	–	Reduced GWP of paddy by 30%–90%
6. Micro-irrigation systems	Jat et al. (2019)	~ 3%–5%	30%	~ 30	~ 30–40	–
7. Subsurface drip irrigation	Jat et al. (2019)		58%–95%	75–169[a]	11–20	
8. Reduction in areas under paddy cultivation and replacing with maize/low ET-demand crop	Gangwar & Singh (2011)	200–450 mm/ha/season	~ 40% as compared to paddy	~ 15	–	Reduced GW use
9. Delayed transplanting of paddy	Punjab and Haryana Subsoil Water Act (2009)	14–90 mm for delay of 1 to 6 weeks	~ 15%–20%	~ 5–10	+ 5–10	Smaller C footprints, reduced GW use

Abbreviations: AWD, alternate wetting and drying; DSWC, Department of Soil & Water Conservation; ET, evapotranspiration; GW, groundwater; GWP, global warming potential; UGPL, underground pipeline.
[a]Energy productivity (Jat et al., 2019).

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Despite AWD's benefits and its potential, the adoption of AWD has been limited, perhaps largely due to farmers' apprehensions that this may lead to yield reductions (Carrijo et al., 2017). Thus, alternatives such as "mild AWD" are recommended, where the cycle of unsaturated conditions is limited compared to AWD (Carrijo et al., 2017). A meta-analysis of 56 studies showed that under mild AWD rice yields were not reduced in most cases (Carrijo et al., 2017) since the roots of the rice plants will still be able to take up water from the saturated soil and the perched water in the root zone. Deelstra et al. (2018) reported an increase in water productivity of 0.59 kg/m^3 under AWD over conventional paddy rice (0.22 kg/m^3) because of water saving and better yields in two districts of Telangana in the Krishna River basin, which demonstrates effectiveness of AWD in resilient agriculture under water deficit conditions. Raising farmers' awareness through extension and capacity building by experts, along with visits to AWD demonstration sites should help to build confidence in scaling up AWD.

Optimizing irrigation scheduling

Irrigation scheduling helps to optimize the application of the right amount of water at the right time and place to optimize crop production, conserve water, and improve performance of irrigation systems. The major finding from a comprehensive review by Sapkota et al. (2020) suggests that optimizing irrigation schedules may help to reduce methane (CH_4) emissions and net GWP. With appropriate irrigation scheduling, optimum soil water conditions for plant growth are maintained and a substantial amount of water is saved. Proper irrigation scheduling also leads to efficient use of fertilizers, thus reducing GHG emissions as well. In India, region-specific irrigation scheduling for major crops has been documented based on field experiments under ICAR-AICRP on water management for surface irrigation and micro-irrigation (All India Coordinated Research Project on Irrigation Water Management (AICRP-IWM), 2019).

There have also been experimental studies on evapotranspiration (ET)-based irrigation scheduling based both on measured actual ET and computed ET (Hunsaker et al., 2015). Soil moisture-based irrigation scheduling is ideal but has not been in practice for many reasons, key among them the cost and lack of farmers' awareness and education. A low-cost irrigation scheduling alternative was demonstrated in Punjab using low-cost tensiometers to measure soil moisture in farmers' fields to optimize rice and wheat irrigation. The findings showed encouraging results with a reduced number of irrigations in paddy cultivation, resulting in about 13% water savings (Vatta et al., 2018). A recent study by Schmitter et al. (2017) in an irrigation command area in Ethiopia demonstrated that building local water user group-based information platforms can support farmers in improving crop and water productivity. The results of their study showed that farmers who have access to information reduced their water application similarly to those who had installed soil moisture sensors (wetting front detectors and chameleon sensors). This can be replicated in India to disseminate advisories to farmers in order to optimize water and nutrient use.

With the recent technological advances in instrumentation, automation, sensor technology, satellite data, artificial intelligence, machine learning, and big data, efforts have been made to develop these technologies for irrigation scheduling and precision farming (Talaviya et al., 2020). They have the potential to provide farmers with enough information to improve irrigation decisions, but access to these alternatives is challenging for small-scale farmers in developing countries because of insufficient network facilities (e.g., information and technology platforms) and/or availability of the technology.

The impact of AWM interventions through reduced consumptive water use, irrigation water saving, yield improvement, and energy saving, along with additional environmental benefits such as GHG emissions reduction, according to available studies, is summarized in Table 1.

Conservation agriculture

CA promotes minimal soil disturbance and maintenance of soil cover through no-tillage, adequate retention of crop residues on the soil surface for mulching, innovative cropping systems, and measures to reduce soil compaction (Jat et al., 2019a, 2019b, 2020). CA-based management practices have been extensively researched and piloted and also implemented at scale in the water- and resource-intensive rice–wheat irrigated north-western states. Results show that CA approaches lead to improvement in crop yield, water saving, water productivity, and net returns in different cropping systems (Jat et al., 2019a, 2019b, 2020). A meta-analysis of CA-based practices in South Asia shows a mean yield advantage of 5.8%, with 12.6% higher water use efficiency, an increase in net economic returns of 25.9%, and a reduction of 12%–33% in GWP (Jat et al., 2020). Studies on layering various management practices by Kakraliya et al. (2018) and bundling SDI with CA, as CA plus, in a rice–wheat (RW) rotation in the western Indo-Gangetic plains (Table 2) by Jat et al. (2019b) reveal its significance in building resilience to climate change. The field-based evidence in Table 2 clearly suggest that CA contributes to all the three key elements of CSA (productivity, resilience/adaptation, and mitigation).

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TABLE 2 Impact of CA-based management practices on irrigation water use, water productivity, net profit, and global warming potential

Management practice	Reference	Water saving	WUE/WP	Yield change	Net income	Reduction in global warming	Other change
CA-based management practices	Jat et al. (2020)	–	+5.8% (WUE)	+12.6%	+25.9%	−12% to −33%	
CA in rice–wheat rotation system	Kakraliya et al. (2018)	+22%	WP (irrigation) + 37% WP (irrigation + rain) by +26%	–	+19%	−40%	System productivity by +6%
CA with subsurface drip irrigation (CA+)	Jat et al. (2019b)	58.4%–95.5%	+145% (WP)	+11.2	+29.2%	–	Energy productivity by +75%–169%

Abbreviations: CA, conservation agriculture; WP, water productivity; WUE, water use efficiency.

Broad bed and furrow system

The broad bed and furrow (BBF) system is an improved in situ soil and moisture conservation and drainage technology for clayey soils with low infiltration rates as the soil gets saturated and waterlogged with the progression of the rainy season (El-Swaify et al., 1985). The BBF system consists of a relatively flat bed or ridge approximately 105 cm wide and a 45-cm-wide shallow furrow (with a 15 cm depth). The BBF system is laid out on a grade of 0.4%–0.8% for optimum performance and is recommended for the deep black soils of the semi-arid region (Anantha et al., 2020). The benefits of BBF include improved yields, better drainage of excess water, soil moisture conservation, and reduced CO_2 emissions (Anantha et al., 2020; Sikka et al., 2018; Singh et al., 2009). On-farm trials in the Vertisols of central India revealed that the BBF system resulted in an average 18% increase in the yields of soybean, maize, chickpea, green gram, pearl millet, and pigeon pea and a 30% increase in rainfall use efficiency (Singh et al., 2009). BBF is being widely promoted as a potential climate-resilient technology in the semi-arid areas of Maharashtra, Madhya Pradesh, Karnataka, Uttar Pradesh, and Rajasthan.

Water productivity in canal command areas

Improving agricultural water productivity in canal command areas through AWM interventions, both technical and institutional, can help close the growing gap between the IPC through irrigation projects and their actual utilization. One novel approach to increasing economic water productivity (EWP) in canal command areas has been demonstrated by Amarasinghe et al. (2021) in the water-scarce Sina irrigation system in the semi-arid district of Ahmednagar in Maharashtra to enhance resilience. This approach uses EWP and the water cost curve as tools to reallocate irrigation-consumptive water use and identify economically viable cropping patterns and AWM under different scenarios (normal, dry, and wet years). The results of this study suggest that the cropping patterns that include high-value, water stress-tolerant crops such as orchards (pomegranate, etc.) with micro-irrigation can be a permanent component in the cropping pattern to provide sustainable income even in low-rainfall years. And in moderate-to-good-rainfall years, orchards and fodder in combination with key field crops will increase EWP and the value of output, thus ensuring income resilience.

3.1.2 | Supply augmentation

Limited surface water and over-exploitation of groundwater requires augmentation of water resources through

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rejuvenation of existing infrastructure and tapping additional water resources through rainwater harvesting, managed aquifer recharge, conjunctive water use management, and non-conventional water resources such as saline and alkaline water, wastewater from the municipalities and villages, etc.

Rainwater harvesting for managed aquifer recharge

Soil moisture security and enhanced water storage through water harvesting in surface and subsurface storage structures helps improve resilience at the local level in farming systems. Rainwater harvesting and its judicious utilization through farm ponds and community tanks in rainfed districts demonstrated this ability and enhanced cropping intensity (20%–135%) (Sikka et al., 2018). A recent study of drought-prone watersheds in central India showed that rainwater harvesting through surface water storage (the *haveli* system), check dams, and field infiltration structures led to an increase in groundwater levels by 2–5 m, providing water security in dry years. In addition, crop yields increased by 10%–70%, 20% of fallow lands were brought under cultivation, and overall household income increased by 1740 US $/year on average (Garg et al., 2020). Similarly, the benefits of large-scale adoption of managed aquifer recharge through rainwater harvesting is being widely reported (Patel et al., 2020).

Conjunctive water use

Conjunctive water use refers to the coordinated and planned utilization of surface water and groundwater resources in an integrated manner appreciating interdependence as well as different characteristics of the two water sources. Conjunctive water use management is more effective than focusing on surface water or groundwater in isolation. Irrigation systems should promote conjunctive irrigation of surface and groundwater by design and not by default to reduce the impact of climatic risks on agriculture and income resilience.

One novel way of implementing conjunctive water use management has been successfully piloted in the Gangetic basin, where rainfall is concentrated in 3–4 monsoon months, through an approach known as "Underground Transfer of Floods for Irrigation" (UTFI) (Alam & Pavelic, 2020; Pavelic et al., 2015). UTFI involves strategically installing managed aquifer recharge infrastructure to recharge depleted aquifers with excess wet-season flows that might pose flood risk to a river basin. This both alleviates the flood risk and increases the groundwater available for irrigation (Pavelic et al., 2015). A pilot-scale demonstration of UTFI showed that a single UTFI recharge structure could recharge on average 44,000 m^3 per season during 62–85 days, which is sufficient to irrigate 13 ha of rabi (winter) wheat (Alam et al., 2020a). Global-scale suitability assessments show that at the global scale, areas with high to very high UTFI suitability comprise a crop area of 622 Mha and could provide for approximately 3.8 billion people (Alam & Pavelic, 2020).

4 | PRIORITIZATION OF AWM PRACTICES

AWM interventions are context and situation specific and may or may not work effectively everywhere (Andrieu et al., 2017). Additionally, different AWM interventions have different costs of implementation and varying environmental and economic impacts (Dunnett et al., 2018; Shirsath et al., 2017). Thus, their implementation requires making critical investment decisions that are relevant for current and future scenarios (Dunnett et al., 2018). This requires prioritizing and matching interventions to the local context; otherwise, the effectiveness of the interventions will be reduced (Van Wijk et al., 2020). Further, implementing them also requires an understanding of the trade-offs and synergies between them (Shirsath et al., 2017).

This makes it challenging to identify a portfolio of location-specific, socially relevant, and viable AWM practices to make the right investment decisions. A range of decision support tools and prioritization frameworks have been developed (Andrieu et al., 2017; Brandt et al., 2017; Dunnett et al., 2018; Khatri-Chhetri et al., 2019; Mwongera et al., 2017; Shirsath et al., 2017). These tools and frameworks are mostly developed to assess and identify portfolios of climate-smart agricultural practices (Van Wijk et al., 2020), in which AWM practices form a critical and significant part (Alam & Sikka, 2019). A detailed overview and evaluation of existing tools and frameworks used for prioritization is given by Van Wijk et al. (2020). The existing tools and frameworks can be broadly classified into stakeholder participatory approaches (Andrieu et al., 2017; Khatri-Chhetri et al., 2019), data- and modelling-based approaches (Shirsath et al., 2017; Dunnett et al., 2018), and a mix of both (Brandt et al., 2017; Mwongera et al., 2017). They also have been implemented at scale from field-/farm-based assessments to country-wide assessments (Van Wijk et al., 2020).

The Stakeholder participatory approach integrates local stakeholders' knowledge and experience in prioritizing climate-smart AWM interventions (Andrieu et al., 2017; Khatri-Chhetri et al., 2019). For example, Andrieu et al. (2017) presented the application of the Climate-Smart Agriculture Prioritization Framework

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(CSA-PF) in Mali. The framework consists of four phases: (1) identification of vulnerable areas and farming systems with potentially relevant CSA practices by a committee of experts, (2) identification of best-bet CSA practices by national stakeholders, (3) cost–benefit analysis of best-bet CSA practices, and (4) development of a portfolio and action plan. Stakeholder-based approaches enhance the involvement, interest, and capability of the stakeholders to own and implement the plan developed while helping to include a diversity of stakeholders and their opinions (Khatri-Chhetri et al., 2019). However, a pure stakeholder approach might not be able to factor in the biophysical aspects and resource need of interventions. This is critical as AWM practices are intricately linked to hydrology and alter spatial and temporal water flows (Barron et al., 2009; Calder et al., 2008).

To account for the biophysical and resource availability aspects of interventions, data- and model-based approaches have been used for prioritization (Dunnett et al., 2018; Shirsath et al., 2017). For example, Shirsath et al. (2017) developed a spreadsheet-based methodology that integrates process-based models (crop simulation models), weather generators, and calculators (emission calculators, irrigation requirement calculators, etc.). The tool requires production, economic, and environmental databases of agronomic interventions for different regions and for different climate change scenarios. They applied their tool in 38 districts of Bihar, India, to create a crop technology portfolio. However, data- and model-based approaches can be data intensive, and in data-limited countries (e.g., less developed countries) this may not work. Also, prioritization only based on data- and model-based approaches ignores the local context and barriers for which stakeholder inputs are critical.

Thus, often a combination of the stakeholder and data-/model-based approaches is needed. In the following sections, we present two distinct and complementary approaches for prioritization: one following stakeholder analysis and the other employing a simple water balance-based approach to prioritize smart interventions. The stakeholder-based approach is based on the work developed by Khatri-Chhetri et al. (2019). In contrast to other prioritization frameworks, which only focus on indicators of productivity, adaptation, and mitigation (Andrieu et al., 2017), this also considers implementation feasibility, adoption barriers, and incentive mechanisms since the adoption of interventions is significantly influenced by these indicators. The water balance-based approach presented is a data-/model-based prioritization approach. This approach focuses on the quantitative impact analysis of interventions to prioritize climate-smart interventions (Alam & Sikka, 2019). This approach was developed to present a simple and robust method that has limited data requirements, since current approaches are data intensive and involve subject matter expertise, including the use of crop models and production functions (Dunnett et al., 2018; Shirsath et al., 2017).

4.1 | Application of stakeholder participatory approach

The stakeholder participatory approach developed by Khatri-Chhetri et al. (2019) was used to prioritize interventions to develop district-level climate adaptation plans for resilient farming in India under the project "Strengthening Capacity in India for Scaling-up Climate-Smart Agriculture Technologies, Practices and Services" (ICAR, IWMI, CCAFS, 2020). Here we present the application of the stakeholder prioritization process for the Adilabad District of Telangana, India. The average annual rainfall of the district is 1,146 mm, and 85% of rainfall occurs during the south-west monsoon season (June–September). The district is primarily rainfed, and only 25% of the cultivated area is irrigated. Climate data analysis and stakeholders identified rainfall deficit, dry spells, and extreme rainfall events to be the main risks. Climate data analysis showed 70% of the total district area is drought prone, and 22% of the total district area is prone to extreme rainfall, with dry spells varying in duration from 11 to 24 days. During the stakeholder workshop, the district's climatic risks were ranked on a 0–5 scale, where 0 indicates no risk, 1 is very low risk, 2 is low risk, 3 is medium risk, 4 is high risk, and 5 is very high risk. This ranking is shown in Figure 3. Stakeholders identified dry spells, unseasonal rains, cold waves, heat waves, excess rainfall, floods, and droughts as medium-to-high risks.

A stakeholder consultation workshop was organized in September 2019 to identify and evaluate a range of climate-smart agricultural technologies in the Adilabad District, and 39 stakeholders participated in the workshop. The participants were from the district agriculture

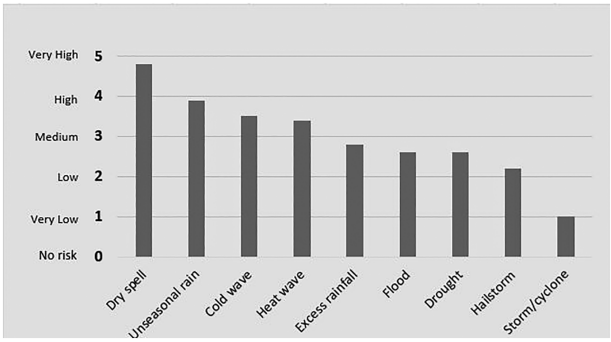


FIGURE 3 Climatic risk ranking marked by stakeholders

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department, extension offices, agricultural research institutions, development organizations, and the private sector (non-governmental organizations, NGOs) along with farmers' representatives and local resource persons. Participants were selected based on their area of work, knowledge of climate adaptation in agriculture, and working experience with farming communities in the district. As a first step, a comprehensive list of CSA technologies (76) categorized as water-smart, energy-smart, carbon-smart, nutrient-smart, weather-smart, and knowledge-smart interventions was shared with local experts and the Krishi Vigyan Kendra (KVK) (Farm Science Centre). They refined this list to exclude or include any irrelevant practices. The refined list of interventions (56) was evaluated for prioritization in the stakeholder workshop.

Stakeholders evaluated CSA interventions on four criteria: (i) their impact on the three CSA pillars of increasing productivity, resilience, and reducing GHG emissions; (ii) implementation feasibility; (iii) adoption barriers; and (iv) incentive mechanisms. Table 3 gives the summary of indicators used for each aspect. To evaluate indicators for the CSA pillars and implementation feasibility, stakeholders used a Likert scale of 0–5 to score interventions where 0 = not relevant, 1 = very low, 2 = low, 3 = medium, 4 = high, and 5 = very high performance. Afterwards, stakeholders' scores on multiple indicators were integrated in a composite climate-smart agriculture performance index (CSA-PI) and technology implementation feasibility index (CSA-IF) based on the relative weightage of each indicator given by stakeholders (Khatri-Chhetri et al., 2019). The CSA-PI and CSA-IF were combined to get a climate-smart score for each AWM practice. This was used to select the prioritized water-smart AWM practices. These prioritized practices were then further evaluated for their adoption barriers and incentive mechanisms to develop policy recommendations. Thus, evaluation of AWM interventions on CSA-PI and CSA-IF along with adoption barriers and incentive mechanisms was used to develop a comprehensive prioritized plan for implementing smart AWM interventions.

TABLE 3 Indicators used for CSA technology prioritization

Evaluation aspect	Indicators
1. CSA pillars	Productivity, income, resilience, and reducing GHG emissions
2. Implementation feasibility	Technical feasibility, cost of technology, gender inclusivity, synergy with government plans
3. Adoption barrier	Availability of finance, inputs, awareness and acceptability of technology, and government support for adoption of technology
4. Incentive mechanism	Available subsidy for technology, access to affordable farm credit, capacity building, and access to market
5. Key institutions to scale out CSA interventions	Private sector retailers, NGOs, farmer producer organizations, custom hiring centres, women self-help groups, young farmer centres

Abbreviations: CSA, climate-smart agriculture; GHG, greenhouse gas; NGO, non-governmental organization.

Prioritized AWM interventions by stakeholders based on the integrated CSA-PI and CSA-IF scores are shown in Figure 4a. The top 10 AWM practices with high climate-smart scores were farm pond, mulching, drip irrigation, stop dams and check dams, farm bunding, sprinkler irrigation, nala bunding, LLL, dug well, and drainage management. Farm ponds and stop/check dams were most preferred and got the highest scores on the technology performance index given their suitability and potential to provide supplemental irrigation in rainfed areas to cope with climate risks. However, the implementation feasibility of farm ponds was rated higher.

Identified key adoption barriers show a significant number of smallholder farmers and their economic situations as the main constraints to technology adoption. Farm pond, drip irrigation, stop dams, and nala bunding (check dams) practices all have funding as the major adoption barrier (Figure 4b). As the majority of farmers lack access to the required inputs, government support and extension services are needed to implement the selected technologies. Rainwater harvesting and labour availability have the least constraints to adoption. There is also a need to raise awareness about the farm pond, mulching, stop dam, LLL, and drainage management technologies. Subsidies were identified as the best incentive to promote and support AWM technologies at a wider scale (Figure 4c). Similarly, capacity building in terms of trainings, agro-advisory support, and demonstration plots was identified as a crucial incentive mechanism for a range of interventions including farm bunding, drainage management, and LLL practices. Affordable credit and linkage with the market were identified as the major incentives to increase LLL adoption.

Stakeholders also identified the critical role of multiple institutions other than the government, such as custom hiring centres, young farmer groups, women's self-help groups, farmer producer organizations, NGOs, and private sector retailers, to expand adoption of prioritized climate-smart AWM practices in the district. The role of

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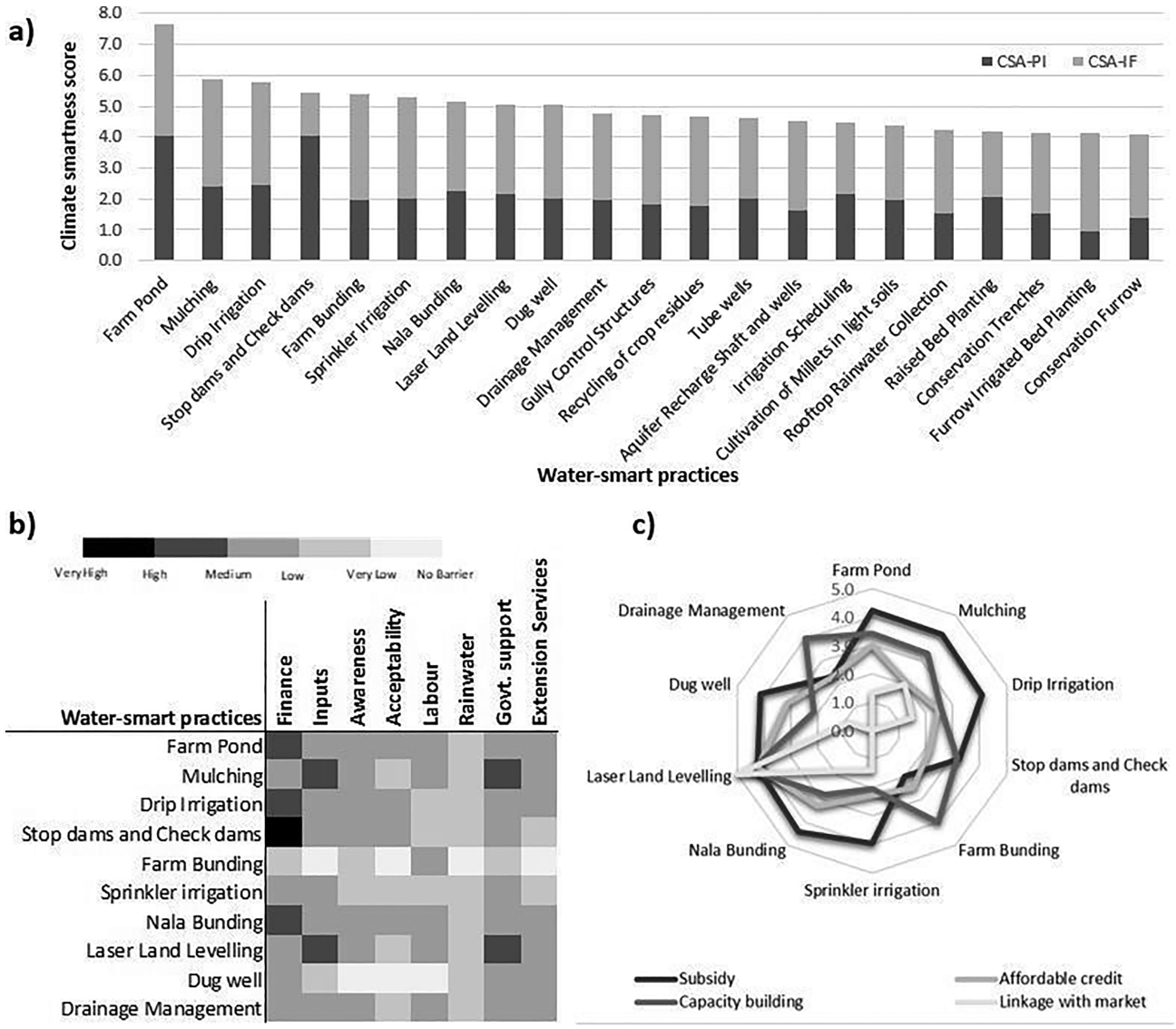


FIGURE 4 (a) Climate smartness score, (b) adoption level and barriers in implementation, and (c) different incentive mechanisms needed for different AWM practices in Adilabad District. AWM, agricultural water management; CSA-PI, climate-smart agriculture performance index; CSA-IF, climate-smart agriculture feasibility index

young farmer groups was identified as being instrumental in promoting technologies at the local level such as drip irrigation, stop dams, nala bunding, dug wells, and LLL. Overall, the stakeholder prioritization framework provided valuable robust information and socially acceptable options for developing a comprehensive investment plan for implementing and expanding climate-smart AWM practices in the district.

4.2 | Application of the water balance approach

The stakeholder prioritization process as discussed above can build a prioritized portfolio of climate-smart AWM practices to a region with incentives and feedback. However, this does not provide quantitative information on their impact on the water balance and cost versus benefits, which can provide further valuable information to help prioritize interventions. This is critical as primarily AWM practices impact crop productivity and resilience through improved moisture/water supply, managed demand, and efficient use of water resources. Not accounting quantitatively for water flow changes or availability can lead to a less-than-optimum portfolio of interventions and even negative externalities (e.g., downstream–upstream impacts).

To evaluate and prioritize AWM practices based on their impact on resource (water) availability, Alam and Sikka (2019) developed in a separate study a simple,

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easily replicable, and conceptually robust water balance tool for implementing interventions. This tool is spreadsheet based and has minimum data requirements to make it easy for users. The tool employs a monthly water balance approach in balancing water resources availability and water demand for given crops. The tool simulates the impacts of AWM interventions on water balance to determine their impact on agricultural productivity, resilience, and mitigation. AWM practices are integrated into the tool by including their expected impact on the physical processes and parameters (e.g., surface storage, groundwater recharge, and runoff) of the water balance. The impact of interventions is quantified across five metrics: change in yield, change in irrigated area, drought resilience, change in GHG emissions intensity, and benefit–cost ratio. The metrics can be compared and combined to select the best portfolio of water-smart interventions based on local context and objectives in consultation with stakeholders.

The tool was implemented in two villages, Barbieri and Sitaljhiri in Sehore, situated in the semi-arid central state of Madhya Pradesh, India. Results showed that different interventions, both individually and combined, show a wide range of impacts and effectiveness in meeting the objectives of CSA as measured by the five metrics. Standalone supply (rainwater harvesting) or demand interventions (sprinkler irrigation) were found to show limited impact on enhanced resilience. This suggests the superiority of combining water supply augmentation with demand and soil and moisture management practices for more resilient agriculture rather than focusing only on standalone interventions. The results were compared with the stakeholder-based approach carried out in the same villages by Khatri-Chhetri et al. (2016). A comparison showed that the results from the two approaches broadly align with an inclination towards the supply-side augmentation measure of rainwater harvesting, the demand-side measure of sprinkler irrigation, and the soil moisture techniques of BBF and conservation tillage (Alam & Sikka, 2019). However, both approaches provided unique insights. The water balance-based approach provided insight on the intensity of the interventions required and their combined impact on achieving climate-smart agricultural objectives. This was not possible with the stakeholder-based approach, which provided only a prioritized list of interventions without information on how much or how to combine them. Similarly, the water balance approach lacked information on adoption barriers, available institutions, and finance which is critical for expanding prioritized practices. This was provided by the stakeholder analysis, which prioritized sprinkler irrigation, with rainwater harvesting also scoring highly on technical feasibility and a high likelihood of attracting support from government subsidy-based schemes. The results and process suggest that the two approaches should possibly be used together to complement each other. The stakeholder participatory process can provide a suitable portfolio of practices obtained through a bottom-up approach, whereas the water balance tool can quantify the impact of these practices on climate-smart metrics (yield, resilience, economic benefits, etc).

5 | THE WAY FORWARD: SCALING UP AWM FOR RESILIENT AGRICULTURE

5.1 | Awareness and capacity building

Expanding smart AWM at scale has been an issue, and awareness and capacity building are the key constraints in improving adaptive capacity to scaling out resilient interventions. An innovative approach to capacity building under the ClimaAdapt Project was implemented from 2012 to 2016 to improve the adaptive capacity of stakeholders and farmers' groups in the Krishna River basin, India (Kakumanu et al., 2019). The study was taken up under the Nagarjuna Sagar Project in the selected command areas in the Nalgonda District of Telangana and the Guntur District of Andhra Pradesh, covering 178 farmers. AWD in rice, a modified system of rice intensification (MSRI) and DSR were the key climate-resilient AWM interventions selected for implementation and capacity building. The impacts of water-saving interventions implemented through capacity building enhanced crop water use efficiency and rice yields. These interventions increased crop yields by 0.96, 0.93, and 0.77 t/ha through AWD, MSRI, and DSR respectively (Kakumanu et al., 2019) and increased farmers' income, clear evidence of enhanced resilience. Village knowledge centres established at the cluster level played a key role in creating awareness about new interventions and providing online services to farmers. This shows the importance for area- and context-specific capacity building of the institutions responsible for water management and of farmers' groups as essential elements in expanding climate-smart AWM practices.

5.2 | Extension services

Experience over the years has shown that in the absence of effective extension services, a sizeable portion of the

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practices used by farmers are derived from traditional wisdom or variants of the technologies. An analysis of 40 years of water management research and outreach in India, using data from 34 centres and 5,000 field trials across 23 states, shows that only 22% of new technologies have been successfully transferred to farmers. The success rate of water management technologies, with the exception of some modern technologies such as LLL and micro-irrigation, is low (Palanisami et al., 2015). In India, the public sector is the major extension service provider, but its reach is constrained because of staff being busy with other non-extension responsibilities (Nandi & Swamikannu, 2019). It is largely the function of the State Department of Agriculture. The Agricultural Technology Management Agency (ATMA) under the National Mission on Agricultural Extension & Technology (NMAET) acts as an umbrella organization for major stakeholders in agriculture and allied activities within a district. ICAR's KVKs perform technology assessments, refinements, and demonstrations and function as knowledge and resource centres, with a vast network of 721 KVKs, through a cooperative effort involving different stakeholders (State Department of Agriculture, research organizations of ICAR, state agricultural universities, and NGOs). For specific crops and regions, some private sector groups and NGOs are also following innovative extension models. The inclusiveness of extension services remains a major concern as the focus of agricultural extension has been on increasing yields, with much less attention on natural resource management. Given that many of the benefits of natural resource management are public goods, the role of the government is critical. Therefore, gaps in technology transfer need to be addressed immediately, with a special emphasis on AWM extension services for scaling out smart AWM interventions at a scale. This should also involve private and civil society extension services as public extension departments alone may not meet the growing service needs of farmers.

5.3 | Convergence

Expanding area-specific water management interventions and production system interventions not only requires coordinated planning across departments/agencies but also vast resources, and this often becomes a constraint. Convergence is now necessary for co-financing in order to provide a critical mass and avoid duplicating works as well as thin distribution of limited resources for meaningful development. The convergence of institutions and co-financing from different development programs to scale up climate-resilient AWM is no exception. Collective planning and implementation between different departments/sectoral institutions will enhance social capital and develop a common understanding, resulting in improved planning, implementation, and management. Through convergence, available resources under different schemes can be targeted in a more meaningful manner to scale up AWM practices for resilient farming. Resources for prioritized AWM interventions can be sourced from existing development programs/schemes, depending on the provisions available and the admissibility of AWM interventions/technologies in those schemes. Scheme-specific convergence guidelines are also available. For instance, the Indian Ministry of Rural Development (MoRD) has developed and disseminated Guidelines for Convergence of the National Rural Employment Guarantee Scheme (NREGS) for different schemes (Ministry of Rural Development (MoRD), 2009).

The United Nations Development Programme (UNDP) and MoRD have documented convergence initiatives and case studies in a joint report (United Nations Development Programme (UNDP) & Ministry of Rural Development, Government of India (MoRD), 2013). Alam et al. (2020b) presented a comprehensive mapping of programmes focusing on different aspects of water management by the government, NGOs, international and national donors, and the private sector in India and illustrated a few examples of convergence models. Meticulously mapping these schemes and their budget allocation shows the potential synergy between their activities which can be converged for effective use of resources in implementing AWM practices for climate-resilient agriculture.

6 | CONCLUSIONS

AWM to enhance water resilience and adaptation to climate change is critical to sustain agriculture, especially in already water-stressed areas. Management of water and land is one of the main pathways for improving yield, enhancing resilience, and in most cases, reducing GHG emissions—the three pillars of CSA. Field-based evidence from a number of studies have demonstrated that water demand and supply management AWM practices and technologies (DSR, LLL, AWD, drip irrigation, CA, subsurface drip, optimized irrigation scheduling, BBF system, water harvesting and aquifer recharge) significantly contributed to enhancing resilience via reduced non-beneficial losses and increasing land and water use efficiency, thereby conserving more water, increasing or maintaining yields, and reducing energy and fertilizer usage, thus also reducing GHG emissions. Prioritization of a location-specific, socially relevant, and viable

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portfolio of smart AWM practices for making the right investment decisions is an important task. We showcase the application of the two distinct but complementary prioritization approaches: stakeholder analysis and a simple water balance-based approach for prioritization of AWM interventions. We suggest that the two approaches should possibly be used together to complement each other. The stakeholder participatory process can provide a suitable portfolio of practices obtained through a bottom-up approach, whereas the water balance model can quantify the impact of these practices on climate-smart metrics (yield, resilience, emissions, economic benefits).

Expanding smart AWM at scale has been an issue, and capacity building, extension services, and the convergence of resources and institutions are among the key constraints. There is evidence to support the impact of capacity building for institutions responsible for water management and farmer groups. This is an essential element in scaling up climate-smart AWM practices. To increase inclusiveness for extension services, it is necessary to involve public, private, and civil society extension services so that farmers' needs can be adequately met. Convergence of institutions and resources for co-financing by sourcing funds for prioritized AWM interventions from relevant development schemes will provide a critical mass for expansion and avoid duplication of work as well as thin distribution of resources. The portfolio of smart AWM interventions and approaches to prioritization of interventions offer promising solutions towards climate-resilient agriculture.

DATA AVAILABILITY STATEMENT

The data that supports the findings of this study are available in the supplementary material of this article.

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